# Day 11: Inference for MLR

## `bikes` data

**Goal:** Predict _____ based on `temp_actual` and `weekend`

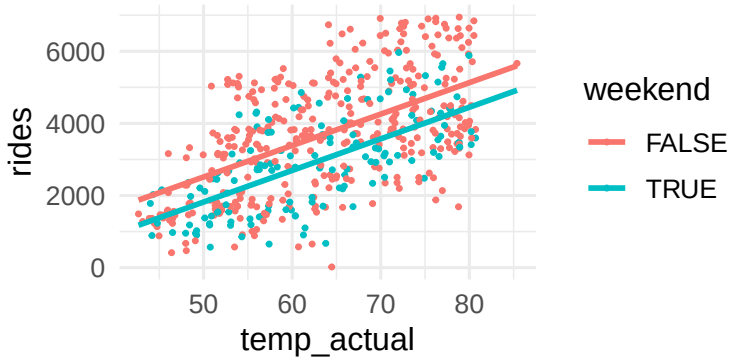
| term | estimate | std. error | t | p-value |
|---|---|---|---|---|
| Intercept | -1843.867 | 415.700 | -4.44 | <0.001 |
| temp_actual | 87.173 | 6.439 | 13.54 | <0.001 |
| weekendTRUE | -714.938 | 795.698 | -0.90 | 0.369 |
| temp_actual:weekendTRUE | 0.411 | 12.539 | 0.03 | 0.974 |

**Warm up:**

Write out the theoretical model from the table above

Write out the fitted model for weekends and weekdays separately

## Do we need a different-slopes model?




**Answer based on the graph:**

**Answer based on the table:**

## T-test for a single coefficient

**Hypotheses:** $H_0 : \beta_j = \#$ vs. $H_a : \beta_j \neq, <, \text{or} > \#$

**Test statistic:** $t = \dfrac{\hat{\beta}_j - \#}{SE(\hat{\beta}_j)}$

**Reference distribution:** $t$ distribution with $df = n - (p + 1)$, where $p$ = number of predictors in the model (an interaction counts as one)

**p-value:** area in the tail(s) specified by $H_a$

**CI for a single coefficient:**

$\hat{\beta}_j \pm t^*_{n-(p+1)} \cdot SE(\hat{\beta}_j)$

**Example:** Use the coefficient table at the top of the notes and `qt(0.975, df = 496)` to calculate a 95% confidence interval for the slope of `temp_actual` (the slope for weekdays).

Do we need weekend at all?

**Answer:**

Now drop only the interaction between weekend and `temp_actual` (the parallel-lines model):

| term | estimate | std. error | t | p-value |
|---|---|---|---|---|
| Intercept | -1850.776 | 357.953 | -5.17 | <0.001 |
| temp_actual | 87.281 | 5.520 | 15.81 | <0.001 |
| weekendTRUE | -689.179 | 124.639 | -5.53 | <0.001 |

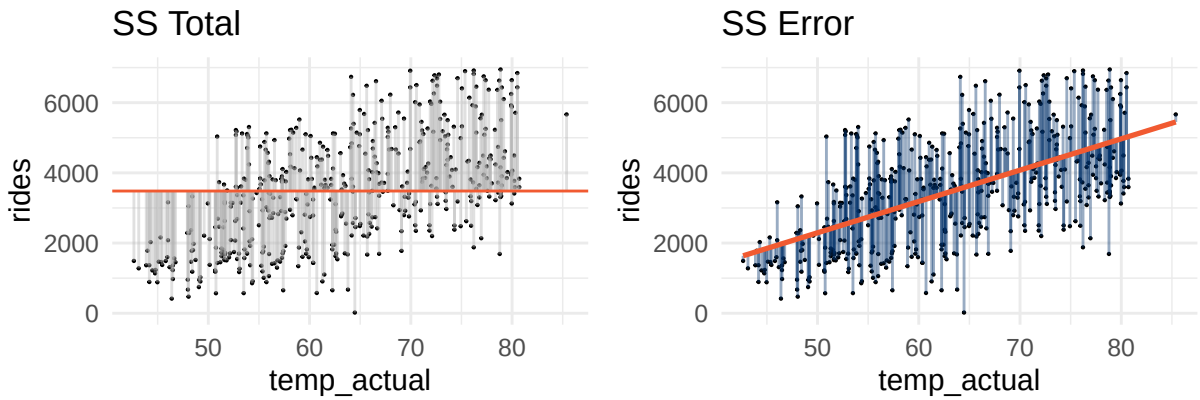
What changed about weekend?

> **Caution!**
>
> $\hat{\beta}$'s and p-values are highly _____ on each other! _____ $\beta$'s might become _____ after removing or adding another term

A t-test can only ask about **one** coefficient at a time, given everything else in the model. We need a test that asks about several coefficients at once

## Explained variability

**Idea:** does the **full** model reduce the error (SSE) by *more than chance alone would*?




## Comparing nested models: the Extra Sum of Squares F Test

Full Model: $\mu\{Y|X\} = \beta_0 + \beta_1 \text{temp} + \beta_2 \text{weekend} + \beta_3 \text{temp} \times \text{weekend}$

Reduced Model: $\mu\{Y|X\} = \beta_0 + \beta_1 \text{temp}$

The reduced model is the full model with some $\beta$'s set to 0, so we say the models are **nested**.

- $H_0$: _____
- $H_a$ : _____

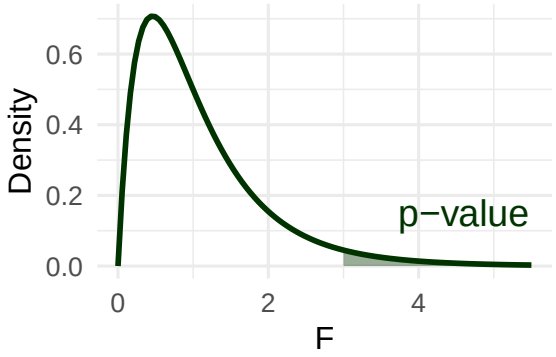
$$F = \frac{(\text{SSE}_{\text{reduced}} - \text{SSE}_{\text{full}})/d}{\text{MSE}_{\text{full}}} = \frac{\text{how much error drops per } \beta}{\text{estimate of } \sigma^2}$$

- If $H_0$ is true, both parts estimate $\sigma^2$, so $F \approx$ _____
- If the extra terms really help, $F$ is much _____ than 1

where

- $d = \text{df}_{\text{reduced}} - \text{df}_{\text{full}}$ is the number of $\beta$'s being tested
- $\text{df}_i = n - (p_i + 1)$ is the error d.f. for model $i$
- $\text{MSE}_{\text{full}} = \text{SSE}_{\text{full}}/\text{df}_{\text{full}}$.

The $F$ statistic follows an $F$ distribution with $d$ and $\text{df}_{\text{full}}$ degrees of freedom. It is right-skewed, and only large values of $F$ are evidence against $H_0$, so the p-value is the **upper tail**:




Upper-tail area in R: `pf(stat, df1, df2, lower.tail = FALSE)`

| Model | SSE | df |
|---|---|---|
| Reduced: temp | 826,326,178 | 498 |
| Full: temp * weekend | 778,436,469 | 496 |

**Example:** Calculate the $F$ statistic.

```
pf(f_stat, df1 = 2, df2 = 496, lower.tail = FALSE)
## [1] 3.703116e-07
```

→ We _____ $H_0$ and conclude:

## Extra Sum of Squares F test in R

```
full <- lm(rides ~ temp_actual * weekend, data = bikes)
reduced <- lm(rides ~ temp_actual, data = bikes)
anova(reduced, full)
```

```
Analysis of Variance Table

Model 1: rides ~ temp_actual
Model 2: rides ~ temp_actual * weekend
  Res.Df       RSS Df Sum of Sq      F    Pr(>F)
1    498 826326178
2    496 778436469  2  47889709 15.257 3.713e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Analysis of Variance Table

Model 1: rides ~ temp_actual
Model 2: rides ~ temp_actual * weekend
  Res.Df       RSS Df    Sum of Sq      F               Pr(>F)
1    df1      SSE1
2    df2      SSE2  d    SSE1-SSE2  (SumSq/d)/(SSE2/df2) p-val
```

## Resolving the puzzle

Earlier, the t-tests in the full model said neither weekend term was useful, but weekend was highly significant once the interaction was dropped. The F-test lets us ask each question directly:

| Question | Reduced | Full | d | F | p-value |
|---|---|---|---|---|---|
| Different slopes needed? | temp + weekend | temp * weekend | 1 | 0.00107 | 0.974 |
| Weekend shift needed (no interaction)? | temp | temp + weekend | 1 | 30.6 | <0.001 |
| Weekend needed at all? | temp | temp * weekend | 2 | 15.3 | <0.001 |

- One tool answers all three questions
- When $d = 1$, the F-test and the t-test are the same test: $F = $ _____
- The t-tests couldn't answer the *joint* question (row __)

## `anova()` on a single model

```
anova(bikes_mlr)
```

```
Analysis of Variance Table

Response: rides
                     Df    Sum Sq   Mean Sq  F value    Pr(>F)
temp_actual           1 411018312 411018312 261.8905 < 2.2e-16 ***
weekend               1  47888023  47888023  30.5130 5.361e-08 ***
temp_actual:weekend   1      1686      1686   0.0011    0.9739
Residuals           496 778436469   1569428
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- Each row adds one term to the model, in order, and tests whether it improves the fit _____ the terms above it
- The **last row** is the extra sums of squares F-test for the last term (here, the interaction) and matches `anova(bikes_mlr2, bikes_mlr)`

## Reading R output

```
lm(formula = rides ~ temp_actual * weekend, data = bikes)

Coefficients:
                         Estimate Std. Error t value Pr(>|t|)
(Intercept)             -1843.867    415.700  -4.436 1.13e-05 ***
temp_actual                87.173      6.439  13.538  < 2e-16 ***
weekendTRUE              -714.938    795.698  -0.899    0.369
temp_actual:weekendTRUE     0.411     12.539   0.033    0.974

Residual standard error: 1253 on 496 degrees of freedom
Multiple R-squared:  0.3709,    Adjusted R-squared:  0.3671
F-statistic: 97.47 on 3 and 496 DF,  p-value: < 2.2e-16
```

The **F-statistic** at the bottom is the same kind of test: the full model vs. an *intercept-only* model.

$H_0$: _____ (none of the predictors help)

## Adjusted $R^2$

$$R^2 = 1 - \frac{SSE}{SST} \qquad\qquad R^2_{adj} = 1 - \frac{SSE/(n-p-1)}{SST/(n-1)} = 1 - \frac{MSE}{MST}$$

where $p$ is the number of predictors in the model.

**Example:** Look at the adjusted $R^2$ column. Which model has the largest adjusted $R^2$? Did adding the interaction help?

| Model | $R^2$ | Adjusted $R^2$ |
|---|---|---|
| temp | 0.3322 | 0.3308 |
| temp + weekend | 0.3709 | 0.3683 |
| temp * weekend | 0.3709 | 0.3671 |
| temp + 5 random noise variables | 0.3392 | 0.3312 |

- $R^2$ **never** goes _____ when predictors are added, even for pure _____
- $R^2_{adj}$ **could** _____ if unnecessary predictors are added
- Better to use $R^2_{adj}$ than $R^2$ to *compare* models, but neither is a formal test

## Takeaways

- A t-test asks about **one** coefficient, *given* everything else in the model
- To test **several** coefficients at once, compare nested models with an **extra sums of squares F-test**
- $F = \dfrac{(\text{SSE}_{\text{reduced}} - \text{SSE}_{\text{full}})/d}{\text{MSE}_{\text{full}}}$: error reduction per extra $\beta$, relative to the noise
- $R^2$ always goes up; the F-test tells us whether the improvement is *real*